

CHANTAL PELLEGRINI

chantal.pellegrini@tum.de
chantalmp.github.io | Google Scholar | GitHub

EDUCATION

PhD Candidate in Computer Science - Technical University of Munich	<i>Nov 2022 - tentatively Nov 2026</i>
Focus: Multimodal foundation models for medical AI	
Advisor: Prof. Nassir Navab	
M.Sc. in Computer Science - Technical University of Munich, Grade: 1.2 (top 10 %)	<i>Apr 2020 - Jul 2022</i>
Exchange Semester - National University of Singapore	<i>Aug 2018 - Dec 2018</i>
B.Sc. in Computer Science - Technical University of Munich, Grade: 1.5 (top 10 %)	<i>Oct 2016 - Mar 2020</i>
studium MINT (Certificate in STEM subjects) - Technical University of Munich	<i>Apr 2016 - Sep 2016</i>
German Abitur - Max-Planck High School Munich	<i>2007 - 2015</i>

EXPERIENCE

Apple	<i>June 2025 - Sep 2025</i>
<i>AI/ML Intern</i>	<i>New York City, USA</i>
• Development of AI-based error detection for Siri Visual responses. • Investigated both zero-shot use of foundation models as well as fine-tuning based approaches. • Deployed final tool as web server and integrated in CI/CD pipeline for automated checks.	

Technical University of Munich	<i>Nov 2022 - tentatively Nov 2026</i>
<i>PhD Candidate</i>	<i>Munich, Germany</i>
• Multimodal patient modeling: Worked on learning from heterogeneous patient data including imaging, free-text reports, laboratory values, bedside monitoring, and treatments, ranging from pretraining on patient population graphs for clinical prediction tasks to forecasting longitudinal patient pathways within a hospital stay [2, 7]. • Large vision-language models for medical environments: Developed domain-specific LVLMs and related methods for radiology and surgery, including a conversational radiology assistant and the first LVLM for surgery understanding with inference-time knowledge integration for unseen tools and interactions [3, 4]. This work received the MICCAI 2024 Best Paper Runner-Up Award. I also worked on reinforcement learning methods for calibrated confidence expression in factual question answering [1]. • Medical image understanding and reporting: Developed methods for fine-grained and interpretable medical image understanding, with a focus on explainable zero-shot X-ray diagnosis and structured radiology reporting via hierarchical visual question answering [5, 6]. • Teaching and supervision: Teaching courses in deep learning for medical imaging and supervising bachelor's and master's theses as well as practical courses.	

Itestra GmbH	<i>Feb 2020 - Aug 2020</i>
<i>Student Researcher</i>	<i>Munich, Germany</i>
• Analysis of the usefulness of source code comments using classical machine learning and deep learning-based approaches in natural language processing. • Development of a Visual Studio Code plugin to show the analysed comment usefulness while coding	

Qualicen GmbH	<i>Feb 2019 - Jan 2020</i>
<i>Software Developer</i>	<i>Munich, Germany</i>
• Backend development for automated requirements analysis.	

- Development of an automated testing framework for the company’s software.

AWARDS AND SERVICES

- Best Paper Runner-Up (2nd place) in MICCAI 2024
- Nominated for the Apple Scholars in AI/ML PhD Fellowship 2024
- Acceptance and participation at the ICVSS 2024 Summer School
- Presentation at the open house of the Bavarian Academy of Sciences and Humanities 2024
- Reviewer for top conferences such as MICCAI, IPMI and ECCV
- Travel Grant for MICCAI 2024 in Marrakesh
- Travel Grant for MICCAI 2023 in Vancouver
- best.in.tum award 2022 - promotional program offered by the Department of Informatics of the TU Munich to the top 2% of their students
- Winner of Bertelsmann Robotics Hackathon 2018
- German Government Scholarship 2017-2018

SKILLS

Programming Languages and Frameworks Languages

Python, PyTorch, Java, MATLAB, C++, OCaml
German (native), English (fluent), Spanish (basic)

SELECTED PUBLICATIONS

- [1] David Bani-Harouni*, **Chantal Pellegrini***, Paul Stangel, Ege Özsoy, Kamilia Zaripova, Matthias Keicher, and Nassir Navab. Rewarding doubt: A reinforcement learning approach to calibrated confidence expression of large language models. *accepted at ICLR 2026*, 2026.
- [2] **Chantal Pellegrini**, Ege Özsoy, David Bani-Harouni, Matthias Keicher, and Nassir Navab. From EHRs to patient pathways: Scalable modeling of longitudinal health trajectories with LLMs. *arXiv preprint arXiv:2506.04831*, 2025.
- [3] **Chantal Pellegrini**, Ege Özsoy, Benjamin Busam, Benedikt Wiestler, Nassir Navab, and Matthias Keicher. Radiolog: Large vision-language models for X-ray reporting and dialog-driven assistance. In *Medical Imaging with Deep Learning*, 2025.
- [4] Ege Özsoy*, **Chantal Pellegrini***, Matthias Keicher, and Nassir Navab. Oracle: Large vision-language models for knowledge-guided holistic OR domain modeling. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 455–465. Springer, 2024.
- [5] **Chantal Pellegrini**, Matthias Keicher, Ege Özsoy, and Nassir Navab. Rad-restruct: A novel vqa benchmark and method for structured radiology reporting. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 409–419. Springer, 2023.
- [6] **Chantal Pellegrini***, Matthias Keicher*, Ege Özsoy*, Petra Jiraskova, Rickmer Braren, and Nassir Navab. Xplainer: From X-ray observations to explainable zero-shot diagnosis. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 420–429. Springer, 2023.
- [7] **Chantal Pellegrini**, Nassir Navab, and Anees Kazi. Unsupervised pre-training of graph transformers on patient population graphs. *Medical Image Analysis*, 89:102895, 2023.
- [8] David Bani-Harouni, **Chantal Pellegrini**, Ege Özsoy, Matthias Keicher, and Nassir Navab. Language agents for hypothesis-driven clinical decision making with reinforcement learning. *accepted at ICLR 2026*, 2026.

- [9] Ege Özsoy, **Chantal Pellegrini**, Tobias Czempel, Felix Tristram, Kun Yuan, David Bani-Harouni, Ulrich Eck, Benjamin Busam, Matthias Keicher, and Nassir Navab. MM-OR: A large multimodal operating room dataset for semantic understanding of high-intensity surgical environments. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 19378–19389, 2025.

** denotes shared first authorship*